

RESEARCH ARTICLE

Learning the Manifold of Authenticity: Hybrid-Curvature Representation Learning for Generalizable Deepfake Detection

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ABSTRACT The practical utility of deepfake detectors is crippled by a crisis of generalization: models that perform well on known manipulation techniques consistently fail when faced with unseen forgeries. We argue this failure stems from a fundamental geometric mismatch. Existing methods implicitly assume that the manifold of authentic faces can be modeled in a space of uniform curvature, typically Euclidean, which inadequately captures the complex, multi-scale structure of facial features. This paper validates the hypothesis that authentic faces lie on a manifold whose geometry is inherently hybrid, requiring both angular compactness (a spherical property) and hierarchical organization (a hyperbolic property). To resolve this geometric mismatch, we introduce a novel detector, CTrue ("ConTrastive learning in opposite-curvature space"), that learns a unified, hybrid-curvature representation of facial authenticity. Trained exclusively on real faces via self-supervised learning, our method simultaneously projects facial embeddings onto two complementary manifolds: a hypersphere to enforce compactness and a hyperbolic space to model the natural feature hierarchy. A single set of mathematically optimal prototypes acts as a "geometric bridge", unifying the learning objectives in both spaces. At inference, a composite score measures an embedding's deviation from this learned manifold. On challenging cross-dataset and cross-manipulation benchmarks, our method achieves competitive generalization under a strictly pristine-only training setting, showing that hybrid-curvature representations provide an effective and data-efficient alternative for deepfake detection.

INDEX TERMS Contrastive learning, generalized deepfake detection, hyperbolic, one-class anomaly detection.

I. INTRODUCTION

Deepfake detection is facing a generalization crisis. While cutting-edge generative models [1], [2], [3] produce forgeries of ever-increasing realism [4], [5], the detectors designed to spot them remain brittle and unreliable. Models trained to high accuracy on existing deepfake datasets [6], [7], [8], [9], [10], [11] consistently fail when confronted with novel manipulation techniques encountered in the wild [12],

a limitation that severely curtails their practical utility. This persistent failure suggests a flaw more fundamental than insufficient training data or suboptimal network architectures.

This paper argues that a fundamental, often overlooked factor contributing to this crisis is a geometric mismatch. The dominant paradigms in deepfake detection, whether based on binary classification, pseudo-fake augmentation, or existing one-class anomaly detection (OC-AD), share a common, unstated flaw: the implicit assumption that the manifold of authentic faces can be adequately modeled in a space of uniform curvature (typically zero for Euclidean or positive

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for spherical). This assumption fundamentally misaligns with the intrinsic geometry of the authenticity manifold, which possesses both compactness and hierarchical structure (faces contain eyes, which contain pupils; mouths contain lips, and so on). Euclidean and purely spherical geometries cannot simultaneously and efficiently capture these complementary geometric properties.

We therefore propose and validate a central hypothesis: **a significant portion of the failure of deepfake detectors to generalize stems from the limitations of learning representations that do not account for the intrinsic hybrid-curvature geometry of the authentic face manifold.**

To address this challenge, we introduce CTrue (“Contrastive learning in opposite-curvature space”), a framework designed specifically to learn a more faithful representation of the “authenticity manifold” as a one-class anomaly detector trained exclusively on pristine images.

From a methodological standpoint, our technical **innovations** focus on the design of this hybrid-curvature representation framework:

- **Dual-Space Embedding Unification:** We design a single-encoder architecture that maps images to a feature space and projects them simultaneously onto a hyperspherical space (to enforce angular compactness and clean boundary separation) and a hyperbolic space (to explicitly model the tree-like, hierarchical structure of facial features).
- **Dual-Optimal Prototypes:** We introduce a novel set of predefined targets that are mathematically optimal in both geometries simultaneously, serving as a structural “geometric bridge” that unifies the learning process across opposite curvatures via a multi-objective loss.
- **Composite Fakeness Scoring:** We formulate a novel inference mechanism that mathematically combines hyperbolic confidence with spherical alignment, establishing a robust, two-factor geometric test for authenticity.

By validating this framework, this work yields several distinct scientific and experimental **contributions**:

- 1) **Conceptualization of Geometric Mismatch:** We formalize and empirically support the “geometric mismatch” between the hybrid-curvature manifold of authentic faces and the uniform-curvature assumptions of prior work, identifying it as a major contributing factor to generalization failure.
- 2) **Empirical Validation of Hybrid OC-AD:** We demonstrate that CTrue can achieve competitive generalization performance on challenging cross-dataset and cross-manipulation benchmarks, including WildDeepfake [12], under a strict, pristine-only training paradigm, suggesting that explicitly modeling the authenticity manifold can reduce the need to chase the infinite variety of possible fakes.
- 3) **Robustness to Structural Degradation:** We provide empirical evidence that the learned hybrid-curvature representations are consistently more resilient to image

corruptions and misalignments than uniform-space baselines, confirming that our framework captures intrinsic properties of facial authenticity rather than superficial artifacts.

The remainder of this paper is organized as follows: Section II provides a thematic review of related work. Section III motivates the adoption of hyperbolic geometry through a hierarchical analysis. Section IV details the proposed CTrue framework, including the dual-space learning objectives. Section V presents a comprehensive experimental evaluation. Section VI discusses the theoretical implications and broader impacts. Finally, Section VII concludes the paper.

II. RELATED WORK

This work intersects with deepfake detection and geometric deep learning. Rather than a simple survey, this section thematically dismantles alternative paradigms to logically motivate the necessity of our hybrid-curvature approach [13], [14].

Numerous deepfake detectors have been proposed in the literature over the years [11], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34].

Theme 1: The Limits of Supervised Classification. The most common approach to deepfake detection is to train a binary classifier to distinguish between real and fake images [11], [35], [36], [37], [38], [39], [40]. These methods have achieved high accuracy on specific datasets but suffer from a documented and persistent failure to generalize to unseen manipulation techniques [6], [12]. This is because they tend to learn the superficial, high-frequency artifacts specific to a particular generation method (e.g., GAN upsampling artifacts) rather than a fundamental concept of authenticity. When a new forgery technique with different artifacts appears, these detectors are easily fooled. This establishes the core problem: accuracy on known fakes is a solved problem; generalization is not.

Theme 2: Incremental Improvements via Data Augmentation. To combat poor generalization, a family of methods has emerged that uses pseudo-fakes, i.e., synthetic forgeries created via data augmentation, to train a more robust binary classifier [41], [42], [43], [44], [45]. Face X-ray [41] blends regions from two different faces, while SBI [42] blends regions from the same face. While these techniques improve generalization to some extent by exposing the model to a wider variety of artifacts, they are fundamentally engaged in a Sisyphean task: trying to guess the appearance of an infinite variety of unseen fakes. This approach treats the symptoms (a lack of diverse training data) rather than a possible underlying factor in poor generalization (learning the wrong underlying representation).

Theme 3: The Promise and Pitfalls of Anomaly Detection. A more principled approach is to reframe the problem as one-class anomaly detection (OC-AD) [25], [46], [47], [48], [49], [50]. Instead of trying to characterize all possible fakes,

OC-AD methods focus on learning a tight, comprehensive representation of the “normal” class, in this case, authentic faces. Anything that deviates significantly from this learned manifold of authenticity is flagged as an anomaly. This is the correct paradigm, as it avoids the need to anticipate future forgery techniques [51], [52], [53], [54], [55].

Beyond digital forensics, our work aligns with trends in advanced anomaly detection across diverse domains. For instance, in hyperspectral image analysis, PUNNet [56] utilizes global feature injection within a blind-spot network to model backgrounds, while NL2Net [57] couples local and non-local features for long-range dependencies. Similarly, in industrial few-shot settings, PCSNet [58] employs prototypical learning to bridge the domain gap. CTrue shares the motivation of modeling underlying distributions through principled constraints but leverages the specific hybrid-curvature geometry of facial manifolds.

However, existing OC-AD methods for deepfake detection have a critical, unstated limitation: their reliance on Euclidean or single-curvature spaces [59]. SeeABLE [59], for instance, uses contrastive learning in a spherical space to detect local discrepancies. While effective, this approach only captures the compactness of the authenticity manifold, ignoring its hierarchical nature. This brings us directly back to our central hypothesis: the geometric mismatch. The promise of OC-AD can only be fully realized if the geometry of the learned representation matches the intrinsic geometry of the data.

Theme 4: The Emergence of Geometric Deep Learning. Hyperbolic geometry, distinguished by its constant negative curvature, has emerged as a powerful tool for modeling hierarchical and tree-like data with minimal distortion [60], [61]. Its ability to embed taxonomies, where the volume of space grows exponentially from the origin, makes it a natural fit for representing semantic relationships [62], [63]. This body of work establishes hyperbolic space as a key enabling technology. By combining it with the complementary properties of spherical space, we can finally build a representation that respects the true geometry of facial authenticity. Few works have explored hyperbolic geometry for deepfake detection (DFD) [19], [23], [64], [65], [66], but they do not explicitly model the manifold of authenticity in a hybrid-curvature space as we propose.

III. MOTIVATION FOR USING HYPERBOLIC GEOMETRY

A. EXPLORING THE HIERARCHICAL STRUCTURE OF DEEPPAKES

Few recent studies have delved into the hierarchical aspects of deepfake-related tasks. In [19], DFD occurs across two hierarchical levels: locally for individual frames and globally through a contrastive paradigm applied to video data spanning multiple frames. Its global objective is the sum of local objectives, computed using a contrastive loss and cross-entropy loss, resulting in a *positive* embedding space. [64] identifies the hierarchical structure in deepfake datasets and classifies manipulated images with fine-grained precision. Its latent space has **null** curvature.

B. CTRUE'S HYPOTHESIS

While the aforementioned works explore the hierarchical structure of deepfakes, they do not *explicitly model it in their latent space*; that is, they do not use a latent space of negative curvature. We hypothesize that the hierarchical structure of deepfakes can be effectively captured by utilizing a latent space of negative curvature. Consequently, we incorporate an additional latent space in the hyperbolic domain, i.e., a Poincaré ball.

Heuristic reasoning: *If DFD is conducted across multiple hierarchical levels, learning in a hierarchical latent space can reveal the inherent structure in the deepfake manifold, enhancing detection performance.*

Instead of relying on explicit labels, whether hierarchical or single-label, CTrue employs a self-supervised learning approach to acquire a hierarchical representation of the data, composed of pseudo-fakes and pristine faces.

The geometric mismatch hypothesis—established in the previous sections as a possible underlying factor of generalization failure in deepfake detection—necessitates a fundamental rethinking of representation learning approaches. Rather than continuing to pursue incremental improvements within single-curvature spaces, we propose a principled solution that directly addresses the core geometric problem: learning representations in a space whose curvature matches that of the authentic face manifold.

To learn the manifold of authenticity, a framework is required that can simultaneously model two essential properties that emerge from the geometric nature of facial data:

- 1) **Angular Compactness:** To ensure that features corresponding to authentic faces are tightly clustered and separable from off-manifold (fake) samples. This property is essential for discriminative separation between authentic and anomalous features.
- 2) **Hierarchical Relationships:** To capture the natural, tree-like structure of facial features (e.g., a face is composed of eyes, a nose, and a mouth; an eye is composed of a pupil, iris, etc.). This property reflects the inherent multi-scale organization of facial components.

These requirements arise naturally from the geometric mismatch hypothesis and immediately justify the need for two complementary geometric spaces. We choose a hypersphere (positive curvature) as the natural and well-established choice for modeling angular compactness, and a hyperbolic space (negative curvature) as the ideal geometry for modeling hierarchical structures. The fusion of these two spaces represents a principled solution to the geometric mismatch problem, directly addressing the theoretical foundation established in the motivation.

C. VALIDATING THE HIERARCHICAL STRUCTURE HYPOTHESIS

We first validate the premise that facial manipulations exhibit a hierarchical structure that can be captured by hyperbolic geometry. We trained an initial version of our

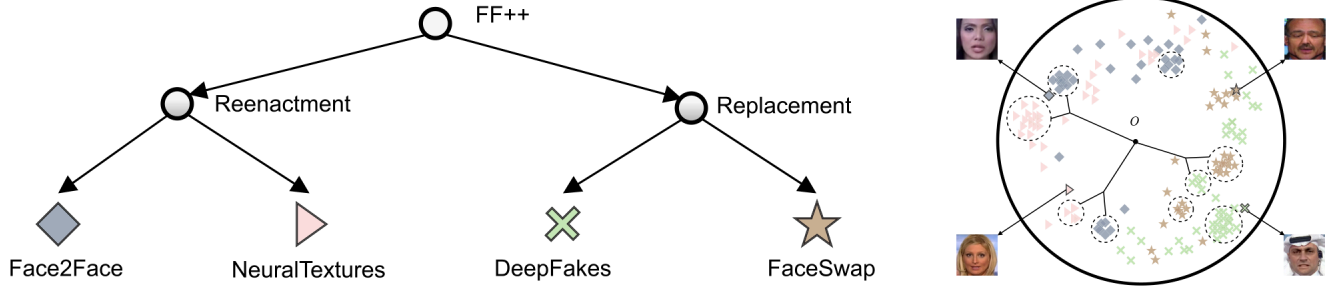


FIGURE 1. (Left) Inherent hierarchical structure of different manipulations in FF++ represented as a tree. **(Right)** t-SNE projection of embeddings for fake images from the FF++ in the Poincaré ball model. Dotted line circles indicate leaf nodes. This visualization provides empirical evidence that a hyperbolic representation naturally captures the hierarchical relationships between different forgery families.

model on pristine faces and pseudo-fakes, then projected the learned hyperbolic embeddings of fake images from the FaceForensics++ (FF++) dataset into a two-dimensional (2D) Poincaré ball using t-distributed stochastic neighbor embedding (t-SNE). As shown in Fig. 1, the visualization reveals a clear hierarchical organization. Broad manipulation categories like ‘Reenactment’ and ‘Replacement’ cluster nearer to the origin, representing coarser semantic concepts. In contrast, finer-grained sub-categories such as ‘Face2Face’ and ‘NeuralTextures’ are pushed further toward the boundary. This empirical observation strongly validates the suitability of hyperbolic geometry for modeling the inherent taxonomy of facial manipulations. Our aim is not to measure this hierarchy, but to leverage a negatively curved latent space to better model the deepfake manifold.

The adoption of hyperbolic geometry is motivated by the inherent hierarchical structure of human facial data. Facial features are naturally organized in a nested tree-like structure, where a global face manifold (root) branches into regional components (eyes, nose, mouth), which in turn decompose into local landmarks and fine-grained textures (leaves). In Euclidean space, the volume of a sphere grows polynomially with the radius, which leads to exponential distortion when embedding such branching structures. In contrast, hyperbolic space possesses an exponential growth of volume that naturally matches the expansion of hierarchical data, providing a strong **geometric inductive bias** for learning low-distortion, multi-scale representations of authenticity.

IV. METHODOLOGY

A constant curvature space is a smooth Riemannian manifold. The curvature sign dictates the space type. A negative curvature represents a hyperbolic space, and we use the Poincaré ball model in this work. A zero curvature implies the Euclidean space, whereas a positive curvature denotes a hypersphere space. Our CTrue (Fig. 2) learns the generic features in spaces of different curvatures through a multi-objective training. The overall procedure of CTrue is summarized in Algorithm 1.

Notation. For simplicity, we denote $\mathbf{x}$ and $\mathbf{y}$ as any two points in a space of positive or negative curvature, with the

Algorithm 1 CTrue Training and Inference

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1: procedure Training( $\mathcal{I}, \mathcal{P}, \eta$ )
2:   repeat
3:      $\mathcal{T}' \leftarrow \text{GeneratePseudoFakes}(\mathcal{I})$ 
4:      $\mathbf{z} \leftarrow f_e(\mathcal{T}'; \theta)$ 
5:      $\mathbf{z}^S \leftarrow \mathbf{z}/\|\mathbf{z}\|; \mathbf{z}^H \leftarrow \exp_0(\mathbf{z})$ 
6:      $\mathcal{L} \leftarrow \mathcal{L}_S(\theta, \mathcal{P}, \mathbf{z}^S) + \lambda \mathcal{L}_H(\theta, \mathcal{P}, \mathbf{z}^H)$ 
7:      $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$ 
8:   until Convergence
9: end procedure
10: procedure Inference( $\mathbf{I}_{test}$ )
11:    $\mathbf{z} \leftarrow f_e(\mathbf{I}_{test}); \mathbf{z}^S \leftarrow \mathbf{z}/\|\mathbf{z}\|; \mathbf{z}^H \leftarrow \exp_0(\mathbf{z})$ 
12:   return  $\max_c [\text{bus}_H(\mathbf{z}^H, \mathbf{p}_c) \times (1 - \text{sim}(\mathbf{z}^S, \mathbf{p}_c))]$ 
13: end procedure

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subscript $_E$ extending to Euclidean space (e.g., $\mathbf{x}_E$ is a point in Euclidean space). We define $\mathbf{z}_i = f_e(\mathbf{I}_i; \theta)$ as the Euclidean embedding produced by the encoder f_e with parameters θ , and $\mathcal{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_C\}$ as the set of shared prototypes acting as class targets. Specifically, in hypersphere space, $\mathbf{x}$ and $\mathbf{y}$ belong to $\mathbb{S}^d$, while in hyperbolic space, they belong to $\mathbb{H}^d$.

A. OVERVIEW OF CTRUE

Given a batch of pristine faces $\{\mathbf{I}_i\}_{i=1}^B$, we generate a batch of pseudo-fakes $\{\mathbf{I}'_i, \tilde{y}_i\}_{i=1}^B$ by applying a data augmentation $\mathbf{I}'_i = \text{psf}(\mathbf{I}_i, \tilde{y}_{loc}, \tilde{y}_{typ})$, where $\tilde{y}_i$ is a self-supervised label corresponding to the specific location and type of the generated anomaly.

The CTrue framework is composed of:

- An encoder $f_e(\theta)$ that maps an image $\mathbf{I}'_i$ to a feature vector $\mathbf{z}_i \in \mathbb{R}^d$.
- Two non-parametric projectors: f_p^S (L2 normalization) for the spherical space $\mathbb{S}^d$, and f_p^H (exponential map) for the hyperbolic space $\mathbb{H}^d$.

A key innovation of our framework is the use of a single set of predefined prototypes $\mathcal{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_C\}$ that serve as a synergistic ‘geometric bridge’ between the two learning objectives. These prototypes are generated as the vertices of a regular simplex, a configuration mathematically proven to be optimal for contrastive learning in spherical space.

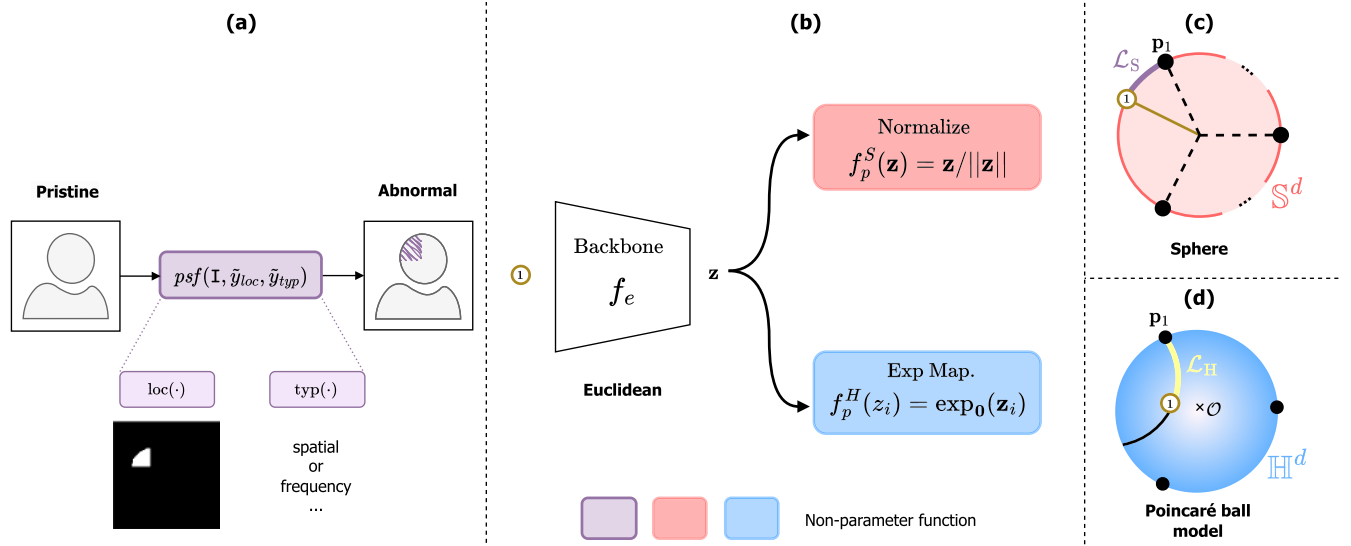


FIGURE 2. Overview of our two-objective training method.(a) Given a pristine face, a pseudo-fake version is generated with a localized anomaly.(b) A single encoder, f_e , produces an embedding z , which is then projected onto the hypersphere S^d and hyperbolic H^d spaces using two non-parametric operations.(c)-(d) The resulting vectors are aligned with their corresponding class prototype using the two learning objectives, $\mathcal{L}_S$ and $\mathcal{L}_H$. The prototypes are shared across both spaces, acting as a geometric bridge.

Crucially, these same points simultaneously correspond to ‘ideal points’ on the boundary of the Poincaré ball, making them perfect targets for the Busemann function in hyperbolic space. This unprecedented dual-optimality is not coincidental but represents the theoretical foundation that enables our unified learning approach. The geometric bridge facilitates a principled, coherent learning process where a single embedding is optimized against a common set of ideal points that possess provable optimal properties in two complementary geometric manifolds. This synergistic design ensures that improvements in one geometric space directly benefit learning in the other, creating a unified representation that is stronger than the sum of its parts.

The loss is a simple sum of the objectives from each space:

$$\mathcal{L}_{C\text{True}} = \mathcal{L}_S(\theta, \mathcal{P}, \{(f_p^S(z_i), \tilde{y}_i)\}_{i=1}^B) + \lambda \cdot \mathcal{L}_H(\theta, \mathcal{P}, \{(f_p^H(z_i), \tilde{y}_i)\}_{i=1}^B) \quad (1)$$

where λ is a balancing hyper-parameter. As we will show in the ablation studies, the inherent complementarity of the two objectives allows us to set $\lambda = 1$ without any tuning.

B. GENERATING ABNORMAL REGIONS

Our self-supervised approach requires generating pseudo-fakes with localized anomalies. We follow the effective blending augmentation strategy from prior work [41], [42]. For a pristine face I , we select a patch at a location $\tilde{y}_{loc}$ from a grid on the face, apply an alteration of type $\tilde{y}_{typ}$, and blend it back onto the original image. The joint location and type $(\tilde{y}_{loc}, \tilde{y}_{typ})$ serves as the self-supervised label for the generated pseudo-fake. We use a 4×4 grid ($N_{loc} = 16$) and two anomaly types ($N_{typ} = 2$: spatial and frequency domain), resulting in $C = 32$ distinct classes.

C. OBJECTIVE 1: LEARNING IN POSITIVELY CURVED SPACE

To enforce angular compactness, we project embeddings onto the unit hypersphere S^d and use a modified supervised contrastive loss [59], [67]. This loss, $\mathcal{L}_S$, encourages embeddings from the same self-supervised class to be pulled closer to each other and to their corresponding predefined class prototype $p_{\tilde{y}_i}$, while being pushed away from embeddings of different classes.

$$\mathcal{L}_S(\theta, \mathcal{P}, \{(z_i^S, \tilde{y}_i)\}_{i=1}^B) = \sum_{i \in \mathcal{I}} \frac{-1}{|\mathcal{P}(i)|} \left(\log \frac{e^{\text{sim}(z_i^S, p_{\tilde{y}_i})}}{\sum_{n \in \mathcal{N}(i)} e^{\text{sim}(z_i^S, z_n^S)}} + \sum_{p \in \mathcal{P}(i)} \log \frac{e^{\text{sim}(z_i^S, p_p^S)}}{\sum_{n \in \mathcal{N}(i)} e^{\text{sim}(z_i^S, z_n^S)}} \right) \quad (2)$$

where sim is the cosine similarity, $\mathcal{P}(i)$ is the set of other samples in the batch with the same label as sample i , and $\mathcal{N}(i)$ contains all other samples. This objective ensures that embeddings of normal data are tightly clustered around well-separated centers in the spherical space.

D. OBJECTIVE 2: LEARNING IN NEGATIVELY CURVED SPACE

Intuitively, the Poincaré ball can be viewed as a space where semantic volume expands exponentially towards the boundary, allowing for a distortion-free embedding of the infinite hierarchy inherent in facial components. To measure proximity in this space, we employ the Busemann function, which can be interpreted as a directional alignment tool. Unlike standard distance metrics, it measures how well an embedding aligns with a ‘prototypical direction’ on the ideal

boundary sphere $\mathbb{S}^d$, effectively acting as a robust confidence score for authenticity.

To enhance our detector's performance by capturing features specific to abnormal faces, we embed our data in the Poincaré ball, leveraging hyperbolic geometry's unique properties [68]. This approach uncovers novel discriminative features that are difficult to capture in Euclidean geometry. In the spherical space, we propel feature vectors projected into the Poincaré ball toward their predefined prototypes (ideal points).

Poincaré ball model is a hyperbolic space defined as:

$$\mathbb{H}_r^d = \{\mathbf{x}_E \in \mathbb{R}^d : \|\mathbf{x}_E\|^2 < r^2, r > 0\}$$

where r represents the radius. In hyperbolic spaces, the traditional vector space formalism does not apply. A gyrovector formalism is therefore introduced [69] to perform operations such as addition. In particular, for a pair of points $\mathbf{x}$ and $\mathbf{y}$ in the space, their Möbius addition is defined as:

$$\mathbf{x} \oplus_r \mathbf{y} = \frac{(1 + 2r^{-2}\langle \mathbf{x}, \mathbf{y} \rangle + r^{-2}\|\mathbf{y}\|^2)\mathbf{x} + (1 - r^{-2}\|\mathbf{x}\|^2)\mathbf{y}}{1 + 2r^{-2}\langle \mathbf{x}, \mathbf{y} \rangle + r^{-4}\|\mathbf{x}\|^2\|\mathbf{y}\|^2}. \quad (3)$$

A remarkable feature of hyperbolic geometry is that the length of circles and the area of disks grow exponentially, in contrast to Euclidean geometry where they grow linearly and quadratically, respectively. In this space, the induced geodesic distance is defined as:

$$\text{dist}_{\mathbb{H}}(\mathbf{x}, \mathbf{y}) = 2r \times \tanh^{-1} \left(r^{-1} \|\mathbf{x} \oplus_r \mathbf{y}\| \right) \quad (4)$$

In hyperbolic geometry, the exponential map $\exp_{\mathbf{x}}^r$ is a function that allows us to map a Euclidean vector to the hyperbolic manifold. More specifically, for a point $\mathbf{x}$ in the hyperbolic space $\mathbb{H}_r^d$ and a vector $\mathbf{v}_E$ in the tangent space $T_{\mathbf{x}}\mathbb{H}_r^d \cong \mathbb{R}^d$, the exponential map is defined as: $\exp_{\mathbf{x}}^r(\mathbf{v}_E) = \mathbf{x} \oplus_r \left(r \tanh \left(\frac{\|\mathbf{v}_E\| \lambda_r(\mathbf{x})}{2r} \right) \frac{\mathbf{v}_E}{\|\mathbf{v}_E\|} \right)$, where $\lambda_r(\mathbf{x}) = \frac{2}{1-r^{-2}\|\mathbf{x}\|^2}$ is the conformal factor in Riemannian geometry.

1) CONNECTION BETWEEN TWO SPACES WITH IDEAL POINTS

In hyperbolic geometry, points at infinity are called *ideal points* and lie on the boundary of the Poincaré ball. Notably, these points share topological properties with the hypersphere $\mathbb{S}_r^d$ [68] (see the following **Section F** for more details). Thus, *our evenly-distributed predefined points can serve as ideal prototypes* for embedding in hyperbolic space. For simplicity, we use a Poincaré ball with $r = 1$ and denote it as $\mathbb{H}^d$ throughout the remainder of this paper.

2) BUSEMANN FUNCTION FOR DETERMINING PROXIMITY TO PROTOTYPES

Measuring the distance between embeddings and prototypes in hyperbolic space accurately is not viable by relying on geodesic distance, as every point $\mathbf{z}_i^H$ in the Poincaré ball $\mathbb{H}^d$ is infinitely distant from any prototypes $\mathbf{p} \in \mathcal{P}$ located on the boundary sphere $\mathbb{S}^d$.

To address this issue, we employ the following function:

$$\text{bus}(\mathbf{z}_i^H, \mathbf{p}) = \lim_{t \rightarrow \infty} \left\{ \text{dist}_{\mathbb{H}}(\gamma_{\mathbf{p}}(t), \mathbf{z}_i^H) - t \right\} \quad (5)$$

where $\gamma_{\mathbf{p}}(t)$ is the geodesic ray starting from the origin $\mathbf{0}$ and ending at the ideal point $\mathbf{p}$ as $t \rightarrow \infty$, and $\mathbf{z}_i^H = \mathbf{f}_p^H(\mathbf{z}_i) = \exp_{\mathbf{0}}(\mathbf{z}_i)$ is the hyperbolic projection of the Euclidean feature from the origin $\mathbf{0}$. Its value can be interpreted as a pseudo-distance, strictly less than 1, indicating the proximity of $\mathbf{z}_i^H$ to the prototype $\mathbf{p}$ on the boundary sphere $\mathbb{S}^d$. Note that $\text{bus}(\mathbf{0}_d, \mathbf{p}) = 0$ and $\text{bus}(\mathbf{z}_i^H, \mathbf{p}) \rightarrow 1$ when $\mathbf{z}_i^H \rightarrow \mathbf{p}$. It can be analytically solved in $\mathbb{H}^d$ (refer to Section F) and has the following closed form: $\text{bus}_{\mathbb{H}}(\mathbf{z}_i^H, \mathbf{p}) = \log(\|\mathbf{p} - \mathbf{z}_i^H\|^2 / (1 - \|\mathbf{z}_i^H\|^2))$.

Our loss function in negatively curved space is defined as the sum of Busemann functions with respect to each embedding and its corresponding class prototype:

$$\mathcal{L}_{\mathbb{H}}(\theta, \mathcal{P}, \{(\mathbf{z}_i^H, \tilde{\mathbf{y}}_i)\}_{i=1}^B) = \sum_{i \in \{1, \dots, B\}} \text{bus}_{\mathbb{H}}(\mathbf{z}_i^H, \mathbf{p}_{\tilde{\mathbf{y}}_i}) \quad (6)$$

By minimizing this loss, we pull embeddings closer to the class prototypes, which correspond to the ideal points at the boundary of the Poincaré ball. In [68], a regularization term was also employed to address the issue of overconfidence, where values close to the ideal boundary were penalized to improve the overall performance. However, in our CTrue model, we conducted experiments and found that the penalization term did not affect DFD performance. This may be because our CTrue model already employs dual losses in two spaces, with the same predefined prototypes. In the spherical space, the non-overlapping loss $\mathcal{L}_{\mathbb{S}}$ does not reach 0, which prevents overconfidence in the hyperbolic space.

E. INFERENCE STAGE: A TWO-FACTOR FAKENESS SCORE

The composite fakeness score is a novel and effective component of our method. It constitutes a highly intuitive two-factor authentication for fakeness. A sample receives a high fakeness score only if it is both **ambiguous (low hyperbolic confidence)** and **misaligned (far from any known normal prototype on the sphere)**. This two-factor check provides the detector with its robustness.

The score combines a confidence term from hyperbolic space with an alignment term from spherical space:

- The **hyperbolic confidence term**, scr_{conf} , leverages a key property of hyperbolic embeddings: generic or ambiguous concepts are mapped near the origin, while specific, well-defined concepts are pushed toward the boundary. This term measures how “well-formed” an embedding is. A pristine face should be near the boundary (high confidence), while a deepfake anomaly should be closer to the origin (low confidence).
- The **spherical alignment term**, $\text{scr}_{\text{align}}$, measures the angular distance between the embedding and its closest “normal” class prototype. A high value indicates the embedding is far from any known configuration of a normal sample.

Formally, for a test image $\mathbf{I}_{test}$, the score is:

$$\text{scr}_{\text{fake}}(\mathbf{I}_{test}) = \max_{c \in \{1, \dots, C\}} \underbrace{\text{bus}_{\mathbb{H}}(\exp_0(f_e(\mathbf{I}_{test})), \mathbf{p}_c)}_{\text{Confidence (scr}_{\text{conf}})}} \times \underbrace{\left(1 - \text{sim}\left(f_p^S(f_e(\mathbf{I}_{test})), \mathbf{p}_c\right)\right)}_{\text{Alignment (scr}_{\text{align}})}} \quad (7)$$

As shown in our ablation study, this multiplicative combination is critical to the method's success.

V. EXPERIMENTS

The experiments are framed not merely as a performance benchmark but as the empirical validation of our geometric mismatch hypothesis.

A. IMPLEMENTATION DETAILS

1) DATASETS AND EVALUATION METRICS

Our model is trained on the *FaceForensics++* (FF++) dataset, as in [43], [44], and [70]. We use only 720 pristine training videos during training and include a maximum of one frame per video when constructing optimization batches. To evaluate CTrue in cross-manipulation settings, we assess its performance on the testing portion of the FF++ dataset. This dataset includes videos generated using four generation techniques: Deepfakes (DF), Face2Face (F2F), NeuralTextures (NT), and FaceSwap (FS). To evaluate the performance of CTrue in cross-dataset scenarios, we mainly use three additional datasets: Celeb-DF-v2 (CDF-v2) [6], DeepFake Detection Challenge preview (DFDC-p) and DFDC public (DFDC) [10]. As our evaluation metric, we adopt Area Under the Receiver Operating Characteristic Curve (AUC) as in prior research [24], [45], [59].

2) DATA PREPROCESSING

For datasets without provided face crops, we perform the following preprocessing steps: (1) we first extract 50 frames from each video and apply RetinaNet [71] to detect the face regions, and then (2) we use Dlib [72] to extract the 68-point facial landmark.

3) THE POSITION OF THE FORGED PATCH

We provide a detailed description of the function $\text{loc}(\cdot)$ which was used in the “grid” masking strategy. This function takes the index $\tilde{y}_{loc} \in \{1, \dots, N_{loc}\}$ as input and generates a binary mask $M_{\tilde{y}_{loc}}$ corresponding to the $\tilde{y}_{loc}$ -th patch of interest using a grid masking strategy.

Given a face image represented as $\mathbf{I} \in [0, 255]^{W \times H \times 3}$, we employ the dlib 68-landmark detector [72] to extract facial landmarks. Subsequently, a rectangle crop is obtained by determining the bounding box that encapsulates all the landmarks. This rectangular region is further divided into a grid consisting of $N_{loc} = N_{rows} \times N_{cols}$ cells. The $\tilde{y}_{loc}$ -th mask $M_{\tilde{y}_{loc}} \in [0, 1]^{W \times H \times 1}$ is then defined as the intersection between the $\tilde{y}_{loc}$ -th region of interest and the convex hull formed by the landmarks.

4) AUGMENTATION

For each image, the global transformations are first applied with the following operations: random translations, random scaling followed by center cropping, and random shifting of HSV channel values. Then for each patch (we use a 4×4 grid, $N_{loc} = 16$), we generate its forged version in both spatial and frequency domains ($N_{typ} = 2$). In the spatial domain, we use random shifting of HSV, RGB channel values, random scaling of the brightness and contrast. In the frequency domain, one of the following operators is used: (1) down-sampling, sharpening/blending filter, and JPEG compression.

5) AUGMENTATION PARAMETERS

The following hyperparameter values were selected in a way that resulted in visually subtle artifacts, similarly to [42]. Specifically, we applied: (i) random scaling (followed by center cropping) by up to 5%, (ii) random shifting of HSV channel values by up to 0.1, and (iii) random translations of up to 3% and 1.5% of the image width and height, respectively. We also use finer-grained augmentations: (i) random shifting of HSV channel values by up to 0.3, (ii) shifting of RGB channel values by up to 20, (iii) random scaling of the brightness and contrast by a factor of up to 0.1, or (iv) down-sampling by a factor of 2 or 4, (v) a sharpening filter and blending with the original with an α value in the range [0.2, 0.5], (vi) JPEG compression with a quality factor between 30 and 70.

6) TRAINING STRATEGY

We employ the AdamW optimizer with an initial learning rate of 0.002 and no weight decay. The model is trained for 200 epochs using the cosine scheduler without restart, preceded by a linear warmup of 20 epochs. We use a batch size of 64. Images are resized to 299×299 before being fed into the model. Training is conducted on two NVIDIA V100 GPUs and requires approximately 16 hours, while during inference, CTrue achieves a frame rate of 74 fps using an RTX 3060. The final CTrue model uses an EfficientNet-B4 backbone and contains 19.3M parameters with 4.4 GFLOPs.

B. CROSS-DATASET PERFORMANCE: VALIDATING GENERALIZATION

Table 1 compares the performance of CTrue to many state-of-the-art competitors in multiple categories: (1) pseudo-deepfake based methods, (2) video-based techniques, (3) transformer-based methods. To make the comparison more explicit, methods are grouped according to their training assumptions, separating methods that rely on fake samples or stronger supervision from pristine-only methods. All models are trained on FF++ (noting that some methods use both pristine and fake videos) and tested on unseen datasets. CTrue demonstrates the best overall performance among pristine-only methods for which all evaluated test sets are reported, despite using only twenty percent of FF++ videos (720 pristine videos from the training set).

TABLE 1. Comparison with state-of-the-art methods in the cross-dataset setting. All models are trained on FF++.

Method	Venue	Test set AUC (%)			
		CDF-v2	DFDC	DFDC-p	Avg. $\uparrow$
<i>Methods using fake samples or stronger supervision</i>					
Two-branch [36]	ECCV'20	76.6	–	–	–
Face X-ray [41]	CVPR'20	79.5	65.5	–	72.5
LipForensics [50]	CVPR'21	82.4	73.5	–	77.9
FTCN-TT [26]	ICCV'21	86.9	74.0	–	80.4
SLADD [43]	CVPR'22	79.7	–	<u>76.0</u>	77.8
Hyperbolic HCIL [19]	ECCV'22	79.0	–	69.2	–
OST [45]	NeurIPS'22	74.8	–	83.3	79.1
TALL + EffNet* [48]	ICCV'23	83.4	67.1	–	–
TALL + ViT-B* [48]	ICCV'23	86.6	74.1	–	–
StyleGRU + SAM [73]	CVPR'24	89.0	–	–	–
ForAda [74]	CVPR'25	<u>92.4</u>	<u>87.2</u>	–	<u>89.8</u>
LOGGER [75]	CVPRW'26	95.9	90.3	–	93.1
<i>Pristine-only methods</i>					
PCL+I2G [44]	ICCV'21	90.0	67.5	74.4	77.3
SBI [42]	CVPR'22	85.9	69.8	74.9	76.9
UIA-ViT [27]	ECCV'22	82.4	–	75.8	79.1
LTTD [24]	NeurIPS'22	89.3	–	80.4	84.8
IIL [47]	CVPR'23	–	–	73.8	–
IIL + SBI	CVPR'23	–	–	79.6	–
IID [76]	CVPR'23	82.0	–	81.2	81.6
SeeABLE [59]	ICCV'23	87.3	<u>75.9</u>	<u>86.3</u>	<u>83.2</u>
SDEQ-Net [77]	Symmetry'26	<u>89.5</u>	–	–	–
CTrue (ours)	–	89.4	77.0	87.9	84.8

Unlike other competitors, CTrue does not require complex multi-head adversarial training schemes or the inclusion of deepfake examples during training. We do not rely on additional information as used in video-based techniques (CTrue operates on a frame-by-frame basis) or employ more powerful transformer backbones. TALL [48] has several versions: CTrue outperforms TALL with EffNet and ViT-B, while slightly underperforming compared to TALL with Swin-T. However, TALL + Swin-T requires 14M additional images from ImageNet-21K and has five times more parameters than our model.

Under this stricter pristine-only protocol, CTrue achieves the best reported performance on DFDC and DFDC-p among pristine-only methods. Recent methods such as LOGGER and ForAda report higher absolute scores on some individual benchmarks, but they operate under a different regime involving stronger supervision, larger pre-trained models, or more complex training pipelines. These results suggest that the proposed dual-curvature representation provides a data-efficient alternative for cross-dataset generalization under a constrained one-class setting.

It should be noted that few works explore hyperbolic geometries for DFD [19], [64], and their methods differ significantly from ours. As shown in Table 1, HCIL [19] performs much worse than our method, while [64] does not report results on these challenging fake-face datasets, instead evaluating on synthetic fake data.

C. CROSS-MANIPULATION PERFORMANCE

As shown in Table 2, CTrue also achieves the best average performance on the FF++ cross-manipulation benchmark. The consistently high AUC across all four manipulation

TABLE 2. Cross-manipulation evaluation on FF++ (c23).

Method	Test set AUC (%)				
	DF	F2F	FS	NT	Avg. $\uparrow$
Face X-ray [41]	–	–	–	–	87.3
SBI [42]	97.5	89.0	96.4	82.8	91.4
OST [45]	–	–	–	–	98.2
SLADD [43]	–	–	–	–	98.4
SeeABLE [59]	<u>99.2</u>	<u>98.8</u>	<u>99.1</u>	<u>96.9</u>	<u>98.5</u>
CTrue (ours)	99.5	99.2	99.5	97.3	98.9

types, despite being trained only on pristine faces, demonstrates that CTrue is not learning to detect specific forgery techniques. Instead, it is effectively identifying deviations from the learned manifold of authenticity, a far more generalizable strategy.

D. ROBUSTNESS ANALYSIS

We evaluated CTrue on contaminated samples using various perturbation methods by adopting the standard evaluation framework proposed in [7], which has been widely used in [24], [49], and [50]. The models were trained on the FF++ raw dataset, and subsequent inference was carried out on the test split, which had been intentionally degraded using diverse perturbations (Table 3).

Faces are cropped and resized before model input, so resizing is not considered corruption. Among frame-based methods like Xception, Face X-ray, and LipForensics, our CTrue shows remarkable robustness to perturbations and achieves superior performance. However, for the specific type of Gaussian noise, two other methods, LTTD and RealForensics, outperform CTrue. This may be due to the fact that these models use additional data during training. For instance, LTTD is a *video transformer-based method* which is more complex than ours, while RealForensics trained the model on *both vision and sound data*, where the sound modality remains unaffected by this type of perturbation, resulting in improved performance.

E. ABLATION STUDIES

1) CONTRIBUTION OF DIFFERENT LOSSES AND SCORES

The ablation study in Table 4 is arguably one of the most important experiments, as it dissects the contribution of each component. Using either the spherical loss ($\mathcal{L}_S$) or the hyperbolic loss ($\mathcal{L}_H$) alone provides reasonable performance, but their combination in CTrue is clearly synergistic, yielding the best results on both datasets.

2) SENSITIVITY TO THE BALANCING HYPERPARAMETER

We further evaluate the sensitivity of CTrue to the balancing coefficient λ in Eq. (1), which controls the relative contribution of the hyperbolic objective. As reported in Table 4, the simple setting $\lambda = 1$ achieves the best performance, with 98.9% AUC on FF++ and 77.0% AUC on DFDC. In comparison, the best fixed alternative λ reaches 97.4% and 74.1%, while a varying λ schedule reaches 98.5% and 75.9%.

TABLE 3. Robustness evaluation under common image corruptions. CS, CC, BW, GNC, GB, and PX denote color saturation, color contrast, block-wise distortion, Gaussian noise, Gaussian blur, and pixelation, respectively. V* and F denote video-based and frame-based methods.

Method	Modality	CS	CC	BW	GNC	GB	PX	Clean $\uparrow$	Avg. $\uparrow$	Drop $\downarrow$
<i>Video-based methods</i>										
LipForensics [50]	V*	99.9	99.6	87.4	73.8	96.1	95.6	99.9	92.1	7.8
LTTD [24]	V*	98.9	<u>96.4</u>	<u>96.1</u>	82.6	97.5	98.6	99.4	<u>95.0</u>	4.4
RealForensics [49]	V*	<u>99.8</u>	99.6	98.9	<u>79.7</u>	95.3	<u>98.4</u>	<u>99.8</u>	95.3	<u>4.5</u>
<i>Frame-based methods</i>										
Xception [35]	F	<u>99.3</u>	98.6	99.7	53.8	60.2	74.2	<u>99.8</u>	81.0	17.8
SBI [42]	F	<u>98.1</u>	96.5	95.3	73.5	78.3	87.9	99.9	88.3	11.6
SeeABLE [59]	F	98.2	96.5	<u>96.1</u>	76.3	82.8	91.1	99.6	<u>90.2</u>	<u>9.4</u>
StyleGRU + SAM [73]	F	99.2	95.8	92.2	55.0	97.3	97.3	99.6	89.5	10.1
CTrue (ours)	F	99.6	<u>97.3</u>	95.2	<u>74.2</u>	<u>89.9</u>	<u>96.8</u>	99.7	92.2	7.5

TABLE 4. Contribution of different loss components. $\mathcal{L}_{SCL}$, $\mathcal{L}_G$, $\mathcal{L}_S$, and $\mathcal{L}_H$ denote the supervised contrastive, geometry-guidance, spherical, and hyperbolic losses, respectively.

$\mathcal{L}_{SCL}$	$\mathcal{L}_G$	$\mathcal{L}_S$	$\mathcal{L}_H$	Scheduler	FF++	DFDC
<i>Single-loss objectives</i>						
✓					91.6	66.8
	✓				51.4	51.6
		✓			94.8	68.5
			✓		91.3	62.3
<i>Combined objectives</i>						
✓	✓			best fixed λ	97.4	74.1
✓	✓			varying λ	<u>98.5</u>	<u>75.9</u>
	✓	✓		$\lambda = 1$	98.9	77.0

These results indicate that CTrue is not sensitive to careful λ tuning and that the two geometric objectives are naturally balanced when equally weighted.

3) ROBUSTNESS TO FACE MISALIGNMENT

We further demonstrate that our method is not reliant on facial symmetry. We evaluated the performance of different loss functions under the intra-dataset FF++ (cross-manipulation) scenarios. The evaluation involved comparing the performance of various losses using an identical set of pristine and manipulated samples. We maintained the images' height while progressively removing a portion of the width (ranging from 5% to 20%) and reported performance on these cropped and resized images. This process was applied symmetrically to both the left and right sides of the face, with the results averaged and presented in Table 5.

CTrue consistently outperformed the baseline SeeABLE performance. While both methods experienced a decline in performance as the crop size increased, our method exhibited a less pronounced drop, highlighting its reduced dependence on facial symmetry. In contrast, SeeABLE's performance showed a significant decline with increasing crop size, which is expected given its reliance on a loss function $\mathcal{L}_G$ that leverages facial geometry.

4) FAKENESS SCORE COMPUTATION

We test different ways of combining the two scores in Eq. 7 to compute the final score. In Table 6, optimal performance

TABLE 5. Robustness to face misalignment.

Loss	Misalign. (%)				
	0	5	10	15	20
$\mathcal{L}_G$ & $\mathcal{L}_S$	<u>98.5</u>	<u>98.0</u>	<u>97.0</u>	<u>95.5</u>	<u>93.1</u>
$\mathcal{L}_H$ & $\mathcal{L}_S$	98.9	98.8	98.5	98.0	97.3

TABLE 6. Fakeness score analysis using confidence and alignment scores.

Score function	AUC (%) $\uparrow$
$\max(s_c) \times \min(s_a)$	96.3
$\min(s_c) \times \max(s_a)$	97.9
$\max(s_c) \times \max(s_a)$	<u>98.3</u>
$\max(s_c \times s_a)$	98.9

TABLE 7. Impact of backbone architecture.

Backbone	FF++
ResNet-50	95.4
Xception	95.8
EfficientNet-B4	98.9

occurs when both scores are maximized simultaneously, as they are complementary. Notably, independent min and max operations lead to degradation due to the interdependence of the two scores. This result is consistent with the observation that the two scores are complementary and should be combined to maximize the overall performance.

5) BACKBONE IMPACT

Table 7 shows the performance of CTrue with several popular CNN backbones. Similar to recent detectors, CTrue works best with EfficientNet-B4 compared to ResNet-50 and Xception.

6) COMPUTATIONAL COMPLEXITY

Table 8 reports the parameter count and GFLOPs of the evaluated backbone variants under the FF++ $\rightarrow$ CDF-v2 setting. FLOPs are reported following the standard configuration of each backbone. The final CTrue model uses an EfficientNet-B4 backbone and contains 19.3M parameters with 4.4 GFLOPs, including the dual-projection heads.

TABLE 8. Computational complexity comparison under the FaceForensics++ → Celeb-DF-v2 setting.

Backbone	Params (M)	GFLOPs ↓	AUC (%) ↑
Xception	22.9	4.8	79.0
Swin-T	28.3	4.5	81.3
EfficientNet-B4 (Ours)	19.3	4.4	89.4

TABLE 9. Additional cross-dataset results on FaceShifter (FSh) and DeeperForensics (DFo).

Method	Venue	FSh	DFo
FTCN	ICCV'21	98.8	98.8
StyleGRU + SAM	CVPR'24	99.0	99.0
CTrue	-	99.1	98.8

TABLE 10. Additional results on FF++ under Raw, high-quality (HQ), and low-quality (LQ) settings.

Method	Raw	HQ	LQ
SLADD	-	98.4	-
SBI	99.9	91.4 (-8.5)	88.1 (-11.8)
SeeABLE	99.6	98.5 (-1.1)	95.8 (-3.8)
CTrue	99.7	98.9 (-0.8)	97.5 (-2.2)

This indicates that the proposed dual-curvature design introduces negligible overhead while maintaining a compact computational footprint.

F. ADDITIONAL RESULTS

1) RESULTS ON FACESHIFTER AND DEEPERFORENSICS

In Table 1, we focused on the very challenging datasets (CDF-v2, DFDC-p, and DFDC-full) rather than older datasets with saturated results like FaceShifter (FSh) [78] and DeeperForensics (DFo) [7]. CTrue is trained only on pristine images, whereas some compared methods use supervised training with fake samples. As shown in Table 9, the results on these older datasets (in the cross-dataset manipulation setting with models trained on the FF++ dataset) demonstrate near-perfect AUC.

2) RESULTS OF INTRA-DATASET LQ SCENARIOS

In Table 2, we evaluate cross-manipulation performance on the FF++ dataset using the HQ (c23) version. SLADD is included as a supervised reference method. Unlike most works that train models on pristine faces and three fake types, then test on a fourth unseen type in a supervised setting, we report results from models trained only on pristine faces and evaluated on all four unseen fake types. Additionally, Table 10 shows CTrue's performance on Raw, HQ, and LQ sets, highlighting its robustness to compression and its superiority over existing methods.

3) CONTRIBUTION OF CURVED SPACES

We further provide visualizations of score distributions for pristine and fake samples obtained from the DFDC dataset

using different scores in Fig. 3. In the left panel, our model trained with $\mathcal{L}_H$ and used scr_{conf} for inference. In the middle panel, the model trained exclusively with $\mathcal{L}_S$ used $\text{scr}_{\text{align}}$ for inference. In the right panel, our model utilizing both spaces shows superior separation, resulting in enhanced performance. The best separation is achieved when using both spaces, as shown in the right panel, which results in enhanced performance. This demonstrates the importance of using both spaces to capture the rich geometric features of deepfake images.

VI. DISCUSSION

A. COMPARISON WITH PREVIOUS WORK

1) RELATION TO DFD USING PSEUDO-FAKES

Unlike other approaches [41], [42], [43], [44] that use data augmentation to create pseudo-fake examples from pristine data and train binary classifiers, we deliberately create localized abnormalities in a controlled manner, without directly aiming to learn the boundary between pristine and pseudo-fake samples.

Although we share the concept of OC-AD, compared to our previous work SeeABLE [59], published at ICCV 2023, CTrue differs in several ways:

- First, SeeABLE uses two losses in spherical spaces, whereas CTrue uses losses in spaces of different curvatures, allowing it to capture richer geometric features of human faces.
- Second, the “guidance” loss in SeeABLE relies on facial geometric constraints as prior knowledge, while CTrue uses two independent and complementary losses that do not require prior knowledge of the image domain. Thus, CTrue is a more versatile model applicable to a broader range of forensic detection tasks.
- Third, SeeABLE requires a specific learning schedule for the two losses, while CTrue does not, simplifying its training process. CTrue outperforms SeeABLE in terms of speed and performance in all evaluated settings. Once trained, CTrue achieves a frame rate of 74 fps on a PC with RTX 3060, while SeeABLE achieves 67 fps.

2) RELATION TO HYPERSPHERICAL AND HYPERBOLIC PROTOTYPE-BASED LEARNING

We highlight here the differences between our approach and hyperspherical prototype networks (HPN) [79] and hyperbolic Busemann-based models [68]. Our innovation lies in the use of a single set of evenly distributed ideal prototypes that are contrastive-optimal in spherical space and ideal-boundary-optimal in hyperbolic space, creating a unique synergistic bridge that simplifies training and enhances generalization.

B. DISCUSSION ON FAIRNESS AND BROADER IMPACT

The architectural design of CTrue offers compelling arguments for improved fairness and broader societal benefits, addressing critical concerns in the field of deepfake detection.

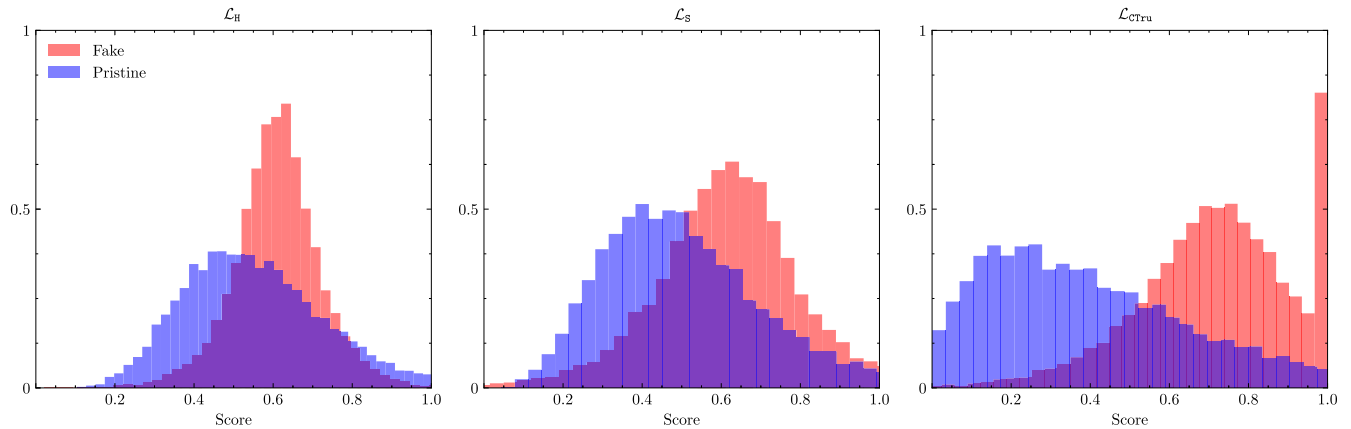


FIGURE 3. Distributions of the normalized scores of pristine samples (blue) and fakes samples (red) from the DFDC dataset.

1) DISCUSSION ON FAIRNESS ISSUES

Our training objectives focus on the position and type of local forged regions. When these positions are identical, *faces of different genders, ages, and ethnicities are linked to the same prototypes*. As a result, we expect our model to exhibit “minimal bias”, or at least less bias than *binary* classification methods. Recent work [76] indicates that identity is a significant factor learned during classifier-based fake detector training.

To address “imbalance” issue, we use *multiple* fixed prototypes in both spaces. This approach aligns with the phenomenon of neural collapse ($\mathcal{NC}$), known for its robustness against imbalance and long-tailed data [80]. We are currently analyzing our method’s fairness and its performance in imbalanced and noisy settings. Additionally, our work benefits society by detecting fake content, including unseen manipulations, thereby fostering greater trust in media.

a: FAIRNESS AND DEMOGRAPHIC BIAS

Traditional binary classifiers trained on real and fake datasets have been shown to inadvertently learn and leverage identity cues, creating known sources of demographic bias [76]. In contrast, our method’s architectural design inherently discourages such bias. Our self-supervised prototypes correspond to *types of local manipulation* (e.g., a patch in the top-left quadrant with a frequency-domain anomaly) rather than facial identities. By training exclusively on authentic faces and using a geometric representation that focuses on deviation from authenticity manifolds rather than identity-specific features, our approach is architecturally discouraged from learning identity-sensitive information. This design principle is a significant advantage over methods that explicitly learn the difference between real fakes faces, which may encode demographic biases present in training datasets.

b: BROADER APPLICATIONS AND SCIENTIFIC IMPACT

The implications of this work extend far beyond deepfake detection. The geometric mismatch hypothesis and the hybrid-curvature solution represent a paradigm shift for

one-class anomaly detection tasks where the normal data class exhibits latent hierarchical structure. Our approach provides a principled foundation for learning representations that respect the intrinsic geometry of the data rather than forcing it into geometrically incompatible spaces.

Potential applications span multiple domains:

- **Medical Imaging:** Detecting subtle, multi-scale anomalies in medical scans (e.g., tumors, lesions), where normal tissue exhibits complex hierarchical organization across multiple scales.
- **Cybersecurity:** Identifying novel network attack patterns by learning the manifold of “normal” network traffic, which exhibits hierarchical organizational properties across protocols and communication layers.
- **Industrial Quality Control:** Finding subtle defects in manufactured goods by modeling the hierarchical structure of a perfectly formed product, where components have natural part-whole relationships.
- **Financial Fraud Detection:** Modeling normal transaction patterns that exhibit hierarchical characteristics (time-based, geographic, merchant-category relationships) to detect anomalous behavior.
- **Scientific Data Analysis:** Detecting anomalous patterns in complex scientific datasets (astronomical images, particle physics data) where normal patterns have inherent hierarchical structure.

c: THEORETICAL CONTRIBUTIONS

Beyond applications, this work contributes to the theoretical understanding of representation learning in anomaly detection. It demonstrates that addressing geometric compatibility between data manifolds and representation spaces is not merely an optimization detail but a fundamental requirement for generalization. The concept of “geometric bridge” prototypes, simultaneously optimal in multiple geometric spaces, opens new theoretical directions for multi-geometry learning.

This positions our contribution as being of foundational importance not just to digital forensics, but to the entire anomaly detection and geometric deep learning communities.

TABLE 11. Representation learning results on CIFAR10 and CIFAR100. $\mathcal{L}_{CE}$ and $\mathcal{L}_{sCL}$ denote cross-entropy and supervised contrastive losses, respectively.

Dataset	$\mathcal{L}_{CE}$	Busemann [68]	$\mathcal{L}_{sCL}^*$	$\mathcal{L}_H$ + ours	$\mathcal{L}_S$	Ours
CIFAR10	94.9	92.3	95.0	94.1	95.2	95.4
CIFAR100	76.1	65.8	76.3	68.0	76.7	77.8

2) PERFORMANCE OF LEARNED REPRESENTATION FOR MULTI-CLASS CLASSIFICATION BEYOND DFD

We further demonstrate the performance of feature representations learned with different losses using CIFAR10/100 with ResNet-18 backbone in Table 11 (considering larger backbones/datasets for future work). The asterisk denotes a two-stage method with linear probing; the other methods are one-stage. Note that CTrue uses only one encoder through multi-objective training and obtains a single embedding, instead of combining two embeddings learned by two encoders as in HCL [81]. Among different considered methods, $\mathcal{L}_{sCL}$ is a two-stage approach: it discards the projection layer and necessitates training an additional linear layer using cross-entropy, while others are one-stage. The method in [68], “ $\mathcal{L}_H$ + our points”, $\mathcal{L}_S$, and “Ours” use cosine similarity between the learned features and prototypes for classification. $\mathcal{L}_H$ with our predefined prototypes outperforms [68] (the advantage of the latter compared to CE (cross entropy) becomes more pronounced when the output features have low dimensions). $\mathcal{L}_S$ surpasses $\mathcal{L}_{sCL}$, and “Ours” performs the best, proving the strength of our learned features in image classification task.

VII. CONCLUSION

This paper confronted the generalization crisis in deepfake detection by addressing what we argue is a major contributing factor: a fundamental geometric mismatch. We have argued and empirically demonstrated that the manifold of facial authenticity is not uniformly curved, and that modeling it with a hybrid-curvature representation can improve generalization. Our proposed detector CTrue realizes this principle by unifying spherical and hyperbolic geometries to learn a more complete and faithful representation of authenticity. Trained only on pristine faces, CTrue achieves competitive performance on challenging generalization benchmarks, particularly among pristine-only methods. These results suggest that hybrid-curvature representation learning is a promising direction not only for deepfake detection, but also for anomaly detection in domains where normal data exhibit latent hierarchical structure.

APPENDIX A

SUMMARY OF NOTATION

Table 12 provides a summary of the primary mathematical symbols and notations used throughout this paper.

TABLE 12. Summary of primary notations.

Symbol	Description
$\mathbf{I}, \mathbf{I}'$	Pristine image and generated pseudo-fake image
f_e, θ	Encoder function and its trainable parameters
$\mathbf{z}$	Euclidean embedding produced by the encoder
$\mathbb{S}^d, \mathbb{H}^d$	Spherical (positive curvature) and hyperbolic (negative curvature) spaces
f_p^S, f_p^H	Non-parametric projection functions for $\mathbb{S}^d$ and $\mathbb{H}^d$
$\mathbf{z}^S, \mathbf{z}^H$	Projected embeddings in $\mathbb{S}^d$ and $\mathbb{H}^d$
$\mathcal{P}, \mathbf{p}_c$	Shared prototype set and prototype associated with class c
$\mathcal{L}_S, \mathcal{L}_H$	Spherical (contrastive) and hyperbolic (Busemann) losses
λ	Hyperparameter balancing the dual-curvature objectives
$\oplus_r$	Möbius addition in the Poincaré ball model
$\exp_0(\cdot)$	Hyperbolic exponential map from the origin
$\text{bus}_H(\cdot)$	Busemann function measuring proximity to the boundary
$\text{sim}(\cdot, \cdot)$	Cosine similarity function
scr_{fake}	Composite fakeness score used during inference
$\text{scr}_{\text{conf}}, \text{scr}_{\text{align}}$	Hyperbolic confidence and spherical alignment scores

REFERENCES

- [1] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, “Generative adversarial networks,” in *Proc. NeurIPS*, vol. 63, 2020, pp. 139–144.
- [2] J. Ho, “Denoising diffusion probabilistic models,” in *Proc. NeurIPS*, vol. 33, 2024, pp. 6840–6851.
- [3] L. Dinh, “Density estimation using real NVP,” in *Proc. ICLR*, 2016.
- [4] P. Oltermann. (2022). *European Politicians Duped Into Deepfake Video Calls With Mayor of Kyiv*. [Online]. Available: <https://www.theguardian.com/world/2022/jun/25/european-leaders-deepfake-video-calls-mayor-of-kyiv-vitali-klitschko>
- [5] S. M. A. K. Chowdhury and J. I. Lubna, “Review on deep fake: A looming technological threat,” in *Proc. 11th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2020, pp. 1–7.
- [6] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, “Celeb-DF: A large-scale challenging dataset for DeepFake forensics,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3204–3213.
- [7] L. Jiang, R. Li, W. Wu, C. Qian, and C. C. Loy, “DeeperForensics-1.0: A large-scale dataset for real-world face forgery detection,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2886–2895.
- [8] P. Korshunov and S. Marcel, “Deepfakes: A new threat to face recognition? Assessment and detection,” Tech. Rep., 2018.
- [9] B. Dolhansky, R. Howes, B. Pflaum, N. Baram, and C. C. Ferrer, “The deepfake detection challenge (DFDC) preview dataset,” 2019, *arXiv:1910.08854*.
- [10] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. C. Ferrer, “The DeepFake detection challenge (DFDC) dataset,” 2020, *arXiv:2006.07397*.
- [11] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Niessner, “FaceForensics++: Learning to detect manipulated facial images,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1–11.
- [12] B. Zi, M. Chang, J. Chen, X. Ma, and Y.-G. Jiang, “Wild deep fake: A challenging real-world dataset for deepfake detection,” in *Proc. ACM Int. Conf. Multimedia*, 2021.
- [13] D. Cozzolino, G. Poggi, and L. Verdoliva, “Recasting residual-based local descriptors as convolutional neural networks: An application to image forgery detection,” in *Proc. 5th ACM Workshop Inf. Hiding Multimedia Secur.*, Jun. 2017, pp. 159–164, doi: [10.1145/3082031.3083247](https://doi.org/10.1145/3082031.3083247).
- [14] X. Yang, Y. Li, and S. Lyu, “Exposing deep fakes using inconsistent head poses,” in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2019, pp. 8261–8265.

- [15] R. Tolosana, R. Vera-Rodriguez, J. Fierrez, A. Morales, and J. Ortega-Garcia, "Deepfakes and beyond: A survey of face manipulation and fake detection," *Inf. Fusion*, vol. 64, pp. 131–148, Dec. 2020. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1566253520303110>
- [16] S. Aneja and M. Nießner, "Generalized zero and few-shot transfer for facial forgery detection," 2020, *arXiv:2006.11863*.
- [17] Y. He, N. Yu, M. Keuper, and M. Fritz, "Beyond the spectrum: Detecting deepfakes via re-synthesis," in *Proc. Thirtieth Int. Joint Conf. Artif. Intell.*, Aug. 2021, pp. 2534–2541.
- [18] R. Shao, T. Wu, and Z. Liu, "Detecting and recovering sequential DeepFake manipulation," in *Proc. ECCV*, 2022.
- [19] Z. Gu, T. Yao, Y. Chen, S. Ding, and L. Ma, "Hierarchical contrastive inconsistency learning for deepfake video detection," in *Proc. ECCV*, 2022, pp. 596–613.
- [20] S. Dong, J. Wang, J. Liang, H. Fan, and R. Ji, "Explaining deepfake detection by analysing image matching," in *Proc. ECCV*, 2022.
- [21] H. Liu, X. Li, W. Zhou, Y. Chen, Y. He, H. Xue, W. Zhang, and N. Yu, "Spatial-phase shallow learning: Rethinking face forgery detection in frequency domain," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 772–781.
- [22] Y.-Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao, "Thinking in frequency: Face forgery detection by mining frequency-aware clues," in *Proc. ECCV*, 2020.
- [23] J. Fei, Y. Dai, P. Yu, T. Shen, Z. Xia, and J. Weng, "Learning second order local anomaly for general face forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 20238–20248.
- [24] E. Ding, J. Guan, Z. Hong, C. Quan, J. Wang, Y. Zhao, and H. Zhou, "Delving into sequential patches for deepfake detection," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 4517–4530.
- [25] C. Feng, Z. Chen, and A. Owens, "Self-supervised video forensics by audio-visual anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10491–10503.
- [26] Y. Zheng, J. Bao, D. Chen, M. Zeng, and F. Wen, "Exploring temporal coherence for more general video face forgery detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15024–15034.
- [27] W. Zhuang, Q. Chu, Z. Tan, Q. Liu, H. Yuan, C. Miao, Z. Luo, and N. Yu, "UIA-ViT: Unsupervised inconsistency-aware method based on vision transformer for face forgery detection," in *Proc. ECCV*, 2022.
- [28] M. Kim, S. Tariq, and S. S. Woo, "FReTAL: Generalizing deepfake detection using knowledge distillation and representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1001–1012.
- [29] C. Li, Z. Huang, D. P. Paudel, Y. Wang, M. Shahbazi, X. Hong, and L. Van Gool, "A continual deepfake detection benchmark: Dataset, methods, and essentials," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 1339–1349.
- [30] (2022). *DeepFakes*. [Online]. Available: <https://www.github.com/deepfakes/faceswap>
- [31] (2022). *FaceSwap*. [Online]. Available: <https://www.github.com/deepfakes/faceswap>
- [32] J. Thies, M. Zollhöfer, M. Stamminger, C. Theobalt, and M. Nießner, "Face2Face: real-time face capture and reenactment of RGB videos," *Commun. ACM*, 2020.
- [33] J. Thies, M. Zollhöfer, and M. Nießner, "Deferred neural rendering: Image synthesis using neural textures," *ACM Trans. Graph.*, vol. 38, no. 4, pp. 1–12, Aug. 2019.
- [34] M. Du, S. Pentyala, Y. Li, and X. Hu, "Towards generalizable deepfake detection with locality-aware AutoEncoder," in *Proc. 29th ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2020, pp. 325–334.
- [35] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1800–1807.
- [36] I. Masi, A. Killekar, R. M. Mascarenhas, S. P. Gurudatt, and W. AbdAlmageed, "Two-branch recurrent network for isolating deepfakes in videos," in *Proc. ECCV*, 2020.
- [37] D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen, "MesoNet: A compact facial video forgery detection network," in *Proc. IEEE Int. Workshop Inf. Forensics Security (WIFS)*, Dec. 2018, pp. 1–7.
- [38] Y. Li and S. Lyu, "Exposing DeepFake videos by detecting face warping artifacts," in *Proc. CVPR Workshop*, 2018.
- [39] H. Dang, F. Liu, J. Stehouwer, X. Liu, and A. K. Jain, "On the detection of digital face manipulation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5780–5789.
- [40] C.-C. Hsu, Y.-X. Zhuang, and C.-Y. Lee, "Deep fake image detection based on pairwise learning," *Appl. Sci.*, vol. 10, no. 1, p. 370, Jan. 2020. [Online]. Available: <https://www.mdpi.com/2076-3417/10/1/370>
- [41] L. Li, J. Bao, T. Zhang, H. Yang, D. Chen, F. Wen, and B. Guo, "Face X-ray for more general face forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5000–5009.
- [42] K. Shiohara and T. Yamasaki, "Detecting deepfakes with self-blended images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 18699–18708.
- [43] L. Chen, Y. Zhang, Y. Song, L. Liu, and J. Wang, "Self-supervised learning of adversarial example: Towards good generalizations for deepfake detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 18689–18698.
- [44] T. Zhao, X. Xu, M. Xu, H. Ding, Y. Xiong, and W. Xia, "Learning self-consistency for deepfake detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15003–15013.
- [45] L. Chen, L. Liu, Y. Song, J. Wang, and Y. Zhang, "OST: Improving generalization of DeepFake detection via one-shot test-time training," in *Proc. NeurIPS*, 2022, pp. 24597–24610. [Online]. Available: <https://proceedings.neurips.cc/paperfiles/paper/2022/file/9bf0810a4a1597a36d27ceea58667d92-Paper-Conference.pdf>
- [46] H. Khalid and S. S. Woo, "OC-FakeDect: Classifying deepfakes using one-class variational autoencoder," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 2794–2803.
- [47] S. Dong, J. Wang, R. Ji, J. Liang, H. Fan, and Z. Ge, "Implicit identity leakage: The stumbling block to improving deepfa," Tech. Rep.
- [48] Y. Xu, J. Liang, G. Jia, Z. Yang, Y. Zhang, and R. He, "TALL: Thumbnail layout for deepfake video detection," in *Proc. ICCV*, 2023, pp. 22601–22611.
- [49] A. Haliassos, R. Mira, S. Petridis, and M. Pantic, "Leveraging real talking faces via self-supervision for robust forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14930–14942.
- [50] A. Haliassos, K. Vougioukas, S. Petridis, and M. Pantic, "Lips Don't lie: A generalisable and robust approach to face forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5037–5047.
- [51] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. ICML*, 2020.
- [52] F. Wang and H. Liu, "Understanding the behaviour of contrastive loss," in *Proc. CVPR*, 2021, pp. 2495–2504.
- [53] F. Gräf, C. D. Hofer, M. Niethammer, and R. Kwitt, "Dissecting supervised contrastive learning," in *Proc. ICML*, 2021.
- [54] P. Awasthi, N. Dikkala, and P. Kamath, "Do more negative samples necessarily hurt in contrastive learning?" in *Proc. ICML*, 2022.
- [55] K. Sohn, C. Li, J. Yoon, M. Jin, and T. Pfister, "Learning and evaluating representations for deep one-class classification," in *Proc. ICLR*, 2020.
- [56] D. Wang, L. Zhuang, L. Gao, X. Sun, and X. Zhao, "Global feature-injected blind-spot network for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
- [57] D. Wang, L. Ren, X. Sun, L. Gao, and J. Chanussot, "Nonlocal and local feature-coupled self-supervised network for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 6981–6993, 2025.
- [58] Y. Jiang, Y. Cao, and W. Shen, "Prototypical learning guided context-aware segmentation network for few-shot anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 12016–12026.
- [59] N. Larue, N.-S. Vu, V. Struc, P. Peer, and V. Christophides, "SeeABLE: Soft discrepancies and bounded contrastive learning for exposing deep-fakes," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 20954–20964.
- [60] M. Nickel and D. Kiela, "Poincaré embeddings for learning hierarchical representations," in *Proc. NeurIPS*, 2017.
- [61] V. Khulkov, L. Mirvakhabova, E. Ustinova, I. Oseledets, and V. Lempitsky, "Hyperbolic image embeddings," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6417–6427.
- [62] O.-E. Ganea, G. Bécigneul, and T. Hofmann, "Hyperbolic neural networks," in *Proc. NeurIPS*, vol. 31, 2018, pp. 5345–5355.

- [63] I. Balazevic, C. Allen, and T. Hospedales, “Multi-relational poincaré graph embeddings,” in *Proc. NeurIPS*, 2019.
- [64] X. Guo, X. Liu, Z. Ren, S. Grosz, I. Masi, and X. Liu, “Hierarchical fine-grained image forgery detection and localization,” *Tech. Rep.*, 2023.
- [65] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [66] J. Wei, S. Wang, and Q. Huang, “F3Net: Fusion, feedback and focus for salient object detection,” in *Proc. ECCV*, 2019.
- [67] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, “Supervised contrastive learning,” in *Proc. NeurIPS*, vol. 33, 2020, pp. 18661–18673.
- [68] M. G. Atigh, M. Keller-Ressel, and P. Mettes, “Hyperbolic busemann learning with ideal prototypes,” in *Proc. NeurIPS*, vol. 34, 2021.
- [69] T. M. Rassias, “A gyrovector space approach to hyperbolic geometry by abraham A. Ungar,” *J. Geometry Symmetry Phys.*, vol. 18, pp. 93–106, 2010.
- [70] J. Tack, S. Mo, J. Jeong, and J. Shin, “CSI: Novelty detection via contrastive learning on distributionally shifted instances,” in *Proc. NeurIPS*, 2020.
- [71] J. Deng, J. Guo, E. Ververas, I. Kotsia, and S. Zafeiriou, “RetinaFace: Single-shot multi-level face localisation in the wild,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5202–5211.
- [72] D. E. King, “Dlib-ml: A machine learning toolkit,” *J. Mach. Learn. Res.*, vol. 10, no. 60, pp. 1755–1758, 2009.
- [73] J. Choi, T. Kim, Y. Jeong, S. Baek, and J. Choi, “Exploiting style latent flows for generalizing deepfake video detection,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 1133–1143.
- [74] X. Cui, Y. Li, A. Luo, J. Zhou, and J. Dong, “Forensics adapter: Adapting CLIP for generalizable face forgery detection,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 19207–19217.
- [75] F. Wu, D. Lu, M. Yao, X. Xu, and F. Guo, “LOGGER: Local-global ensemble for robust deepfake detection in the wild,” in *Proc. CVPRW*, 2026.
- [76] B. Huang, Z. Wang, J. Yang, J. Ai, Q. Zou, Q. Wang, and D. Ye, “Implicit identity driven deepfake face swapping detection,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 4490–4499.
- [77] R. Zhang, B. Li, and D. Xu, “SDEQ-Net: A deepfake video anomaly detection method integrating stochastic differential equations and hermitian-symmetric quantum representations,” *Symmetry*, vol. 18, no. 2, p. 259, Jan. 2026. [Online]. Available: <https://www.mdpi.com/2073-8994/18/2/259>
- [78] L. Li, J. Bao, H. Yang, D. Chen, and F. Wen, “FaceShifter: Towards high fidelity and occlusion aware face swapping,” in *Proc. CVPR*, 2019.
- [79] P. Mettes, E. van der Pol, and C. G. M. Snoek, “Hyperspherical prototype networks,” in *Proc. NeurIPS*, 2019.
- [80] Y. Yang, L. Xie, S. Chen, X. Li, Z. Lin, and D. Tao, “Do we really need a learnable classifier at the end of deep neural network?” in *Proc. NeurIPS*, 2022.
- [81] S. Ge, S. Mishra, S. Kornblith, C.-L. Li, and D. Jacobs, “Hyperbolic contrastive learning for visual representations beyond objects,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6840–6849.



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